POLY PMNA · DIPLOMA CURRICULUM REV2021

Course 4009 | Semester 4 | Project | Shared handbook

# Minor Project

Plan, execute, document, present and evaluate a feasible diploma-level minor project without losing sight of scope, evidence, safety or academic integrity.

| | |
|---|---|
| Course code | **4009** |
| Revision | **REV2021** |
| Semester | **Semester 4** |
| Type | **Project** |
| Catalogue scope | **26 catalogue programme assignments** |

COURSE ORIENTATION

## How to use this handbook

This page is a structured learning aid for the shared catalogue record. Study the concept, complete the applied activity, retain the evidence, and compare the result with the latest approved programme instructions. Where the official syllabus differs, the official record takes precedence.

**Source and verification note:** The course identity, code, semester and subject type are taken from the tracked POLY PMNA REV2021 catalogue, which indexes the official SITTTR scheme. The official course-content reference is SITTTR course 4009; the scheme index is SITTTR REV2021 diploma syllabus. The direct subject-content page was not machine-readable or returned an unavailable-file response during preparation, so module headings and explanatory teaching material on this page are an original study-handbook structure, not claimed verbatim SITTTR wording. Confirm current hours, marks, credits, outcomes and assessment against the latest official programme record before examination use.

### Learning outcomes

- Explain the central concepts and vocabulary of the subject using clear technical language.
- Apply a controlled project workflow to a diploma-level problem with scope, milestones and acceptance evidence.
- Create a proposal, plan, design or final evidence pack that a supervisor can review.
- Identify assumptions, limitations and common errors, then improve the work using feedback.

### Shared-record context

This code is assigned to 26 catalogue programme assignments in the tracked catalogue. The page therefore focuses on common learning practice and avoids claiming programme-specific technical details that were not verified.

**Study rule:** Do not treat the author-created examples as official examination questions or official mark distribution.

STUDY MAP

## Modules and evidence

**Author-created study map; verify official hours and marks**

| Module | Heading | Purpose | Evidence to retain |
|---|---|---|---|
| 1 | Project identity and a feasible problem | Define what a minor project is, distinguish a real problem from a vague topic, and set a technically achievable scope. | Write a one-page problem brief for a programme-relevant device, application, process or service. |
| 2 | Proposal, feasibility and planning | Turn the problem into an approved proposal with a defensible plan and a realistic schedule. | Prepare a proposal table containing objective, task, owner, evidence, due date and risk response. |
| 3 | Design, development and testing | Use an evidence-driven workflow from design to implementation and test the result against stated requirements. | Create a requirements-to-test traceability matrix with at least six test cases. |
| 4 | Documentation, presentation and evaluation | Communicate the project clearly through a report, demonstration and viva while acknowledging limitations. | Draft a five-minute demonstration script and a viva answer for each major design decision. |
| 5 | Assessment evidence and project closure | Assemble a complete evidence pack that allows a supervisor to evaluate both process and outcome. | Use the closure checklist to verify that every claimed outcome has supporting evidence. |

MODULE 1

# Project identity and a feasible problem

Purpose-led study | Apply + reflect

Define what a minor project is, distinguish a real problem from a vague topic, and set a technically achievable scope.

### Core topics–

- Need statement, users and context
- Objectives written with observable verbs
- Scope, exclusions, constraints, ethics and safety
- Deliverables and acceptance criteria

### Study method

Read the concept, connect it to a diploma-level situation, make a labelled record or diagram, and check the result against the stated outcome. Keep programme-specific technical decisions within the approved project, laboratory or workplace context.

### Applied activity

Write a one-page problem brief for a programme-relevant device, application, process or service.

**Evidence:** Keep the completed activity, a short reflection and any source or test record in your course folder.

#### Common mistakes and corrections−

- Starting with implementation before defining the problem, inputs, outputs or acceptance criteria.
- Making a claim without a source, measurement, test or documented observation.
- Using a template mechanically instead of adapting it to the approved programme context.

**Correction:** state the assumption, choose evidence, perform the check, and record the decision.

#### Self-check−

1. Explain the main idea of this module in your own words.
2. Apply it to a small technical or workplace scenario.
3. Identify one likely failure and describe the evidence that would diagnose it.

MODULE 2

## Proposal, feasibility and planning

Purpose-led study | Apply + reflect

Turn the problem into an approved proposal with a defensible plan and a realistic schedule.

#### Core topics−

- Background and related work
- Functional and non-functional requirements
- Technical, economic and operational feasibility

- Work breakdown structure, milestones and risk register

### Study method

Read the concept, connect it to a diploma-level situation, make a labelled record or diagram, and check the result against the stated outcome. Keep programme-specific technical decisions within the approved project, laboratory or workplace context.

### Applied activity

Prepare a proposal table containing objective, task, owner, evidence, due date and risk response.

**Evidence:** Keep the completed activity, a short reflection and any source or test record in your course folder.

#### Common mistakes and corrections−

- Starting with implementation before defining the problem, inputs, outputs or acceptance criteria.
- Making a claim without a source, measurement, test or documented observation.
- Using a template mechanically instead of adapting it to the approved programme context.

**Correction:** state the assumption, choose evidence, perform the check, and record the decision.

#### Self-check−

1. Explain the main idea of this module in your own words.
2. Apply it to a small technical or workplace scenario.
3. Identify one likely failure and describe the evidence that would diagnose it.

MODULE 3

## Design, development and testing

Purpose-led study

Apply + reflect

Use an evidence-driven workflow from design to implementation and test the result against stated requirements.

**Core topics−**

- Architecture or process diagram
- Design decisions and alternatives
- Versioned implementation and change control
- Test cases, edge cases, results and defect correction

### Study method

Read the concept, connect it to a diploma-level situation, make a labelled record or diagram, and check the result against the stated outcome. Keep programme-specific technical decisions within the approved project, laboratory or workplace context.

### Applied activity

Create a requirements-to-test traceability matrix with at least six test cases.

**Evidence:** Keep the completed activity, a short reflection and any source or test record in your course folder.

**Common mistakes and corrections−**

- Starting with implementation before defining the problem, inputs, outputs or acceptance criteria.
- Making a claim without a source, measurement, test or documented observation.
- Using a template mechanically instead of adapting it to the approved programme context.

**Correction:** state the assumption, choose evidence, perform the check, and record the decision.

**Self-check−**

1. Explain the main idea of this module in your own words.
2. Apply it to a small technical or workplace scenario.

3. Identify one likely failure and describe the evidence that would diagnose it.

MODULE 4

## Documentation, presentation and evaluation

Purpose-led study | Apply + reflect

Communicate the project clearly through a report, demonstration and viva while acknowledging limitations.

### Core topics−

- Report structure and citation practice
- Figures, tables, photographs and appendices
- Demonstration script and presentation design
- Individual contribution, limitations and future work

### Study method

Read the concept, connect it to a diploma-level situation, make a labelled record or diagram, and check the result against the stated outcome. Keep programme-specific technical decisions within the approved project, laboratory or workplace context.

### Applied activity

Draft a five-minute demonstration script and a viva answer for each major design decision.

**Evidence:** Keep the completed activity, a short reflection and any source or test record in your course folder.

### Common mistakes and corrections−

- Starting with implementation before defining the problem, inputs, outputs or acceptance criteria.
- Making a claim without a source, measurement, test or documented observation.

- Using a template mechanically instead of adapting it to the approved programme context.

**Correction:** state the assumption, choose evidence, perform the check, and record the decision.

**Self-check−**

1. Explain the main idea of this module in your own words.
2. Apply it to a small technical or workplace scenario.
3. Identify one likely failure and describe the evidence that would diagnose it.

MODULE 5

## Assessment evidence and project closure

**Purpose-led study** **Apply + reflect**

Assemble a complete evidence pack that allows a supervisor to evaluate both process and outcome.

**Core topics−**

- Proposal approval and progress records
- Working artefact and test evidence
- Final report and presentation
- Reflection, handover and archive

### Study method

Read the concept, connect it to a diploma-level situation, make a labelled record or diagram, and check the result against the stated outcome. Keep programme-specific technical decisions within the approved project, laboratory or workplace context.

### Applied activity

Use the closure checklist to verify that every claimed outcome has supporting evidence.

**Evidence:** Keep the completed activity, a short reflection and any source or test record in your course folder.

### Common mistakes and corrections−

- Starting with implementation before defining the problem, inputs, outputs or acceptance criteria.
- Making a claim without a source, measurement, test or documented observation.
- Using a template mechanically instead of adapting it to the approved programme context.

**Correction:** state the assumption, choose evidence, perform the check, and record the decision.

### Self-check−

1. Explain the main idea of this module in your own words.
2. Apply it to a small technical or workplace scenario.
3. Identify one likely failure and describe the evidence that would diagnose it.

ASSESSMENT PREPARATION

## Evidence, revision and quality gates

### Before submission or examination

1. Confirm the current SITTTR/SBTE title, hours, credits, outcomes and assessment.
2. Define the problem or prompt and state assumptions.
3. Show the method, calculation, algorithm, project record or communication structure.
4. Attach test, source, supervisor, workplace or reflection evidence as applicable.
5. Review clarity, safety, ethics, citations, originality and accessibility.

### Revision checklist

- Can I define the main terms without copying?
- Can I apply the method to a small new situation?

- Can I explain one common error and its correction?
- Can I show evidence for each claimed outcome?
- Can I state the limitations and the next improvement?

```
Evidence record
Subject: 4009 — Minor Project
Task or question:
Assumption or requirement:
Method / design / procedure:
Result or artefact:
Test / source / supervisor evidence:
Reflection and next action:
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SELF-CHECK AND EXAMINATION PRACTICE

## Question bank

### Recall question

Define two important terms from Minor Project and distinguish them.
**Answer guidance:** define the terms, state assumptions, show the method, use a specific example, and cite or retain evidence where appropriate.

### Explain question

Explain why the method in this handbook matters for a diploma student.
**Answer guidance:** define the terms, state assumptions, show the method, use a specific example, and cite or retain evidence where appropriate.

### Apply question

Apply one module method to a realistic technical, project or workplace situation.
**Answer guidance:** define the terms, state assumptions, show the method, use a specific example, and cite or retain evidence where appropriate.

### Evaluate question

Identify a weak approach, state the evidence needed, and propose a defensible improvement.
**Answer guidance:** define the terms, state assumptions, show the method, use a specific example, and cite or retain evidence where appropriate.

### Create question

Design a small artefact, plan, test, report section or presentation that demonstrates the subject outcomes.
**Answer guidance:** define the terms, state assumptions, show the method, use a specific example, and cite or retain evidence where appropriate.

REFERENCES AND GLOSSARY

## Reference trail


1. Official SITTTR REV2021 diploma syllabus catalogue.
2. Official SITTTR course-content reference for code 4009.
3. POLY PMNA Study Hub, for the current revision index and related lesson pages.


### Glossary

**Terms used in this handbook**

| Term | Working meaning |
|---|---|
| Outcome | A capability the learner should demonstrate, not just a topic read. |
| Evidence | A record, artefact, result, source, test or review that supports a claim. |
| Acceptance criterion | A condition used to decide whether a requirement or deliverable is satisfactory. |
| Reflection | A brief evidence-based account of what worked, what did not and what should change. |

Prepared as a POLY PMNA study handbook. Always follow the latest official SITTTR/SBTE instructions for the programme and examination session.